

$$\textcircled{D} T_n(x) = \sum_{k=-n}^n C_k e^{ikx} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) \sum_{k=-n}^n e^{ik(x-t)} dt$$

$$\exists e^{ip} \neq 1 \Leftrightarrow p \neq 2\pi k, k \in \mathbb{Z}$$

$$\textcircled{D} D_n(x) = \sum_{k=-n}^n e^{ikp} = \sum_{k=-n}^n (e^{ip})^k = \frac{e^{-ipn}(1 - e^{ip(2n+1)})}{1 - e^{ip}}$$

$$= \frac{e^{-ipn} - e^{ip(n+1)}}{1 - e^{ip}} \left\{ : e^{ip/2} \right\} =$$

$$= \frac{e^{-ip(n+\frac{1}{2})} - e^{ip(n+\frac{1}{2})}}{e^{-ip/2} - e^{ip/2}} = \frac{\sin(n+\frac{1}{2})p}{\sin p/2}$$

$$\exists e^{ip} = 1, \Rightarrow D_n(p) = 2n+1$$

$$D_n(p) = \begin{cases} \frac{\sin(n+\frac{1}{2})p}{\sin p/2}, & p \neq 2\pi k, k \in \mathbb{Z} \\ 2n+1, & p = 2\pi k, k \in \mathbb{Z} \end{cases}$$

Dnp Φ -ум $D_n(p)$ may эфром Дирхле

① ① $D_n(p) - T = 2\pi$ - нуль

② $D_n(p) - \text{сиг}$

③ $\frac{1}{2\pi} \int_{-\pi}^{\pi} D_n(p) dp = 1$

$$T_n(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_n(x-t) dt$$

④ Теорема ① $f \in R_{loc}(a, b) \Rightarrow$

② $\int_a^b |f| dx < +\infty$

$$\int_a^b f(x) e^{i\lambda x} dx \xrightarrow{|\lambda| \rightarrow +\infty} 0, \lambda \in \mathbb{R}$$

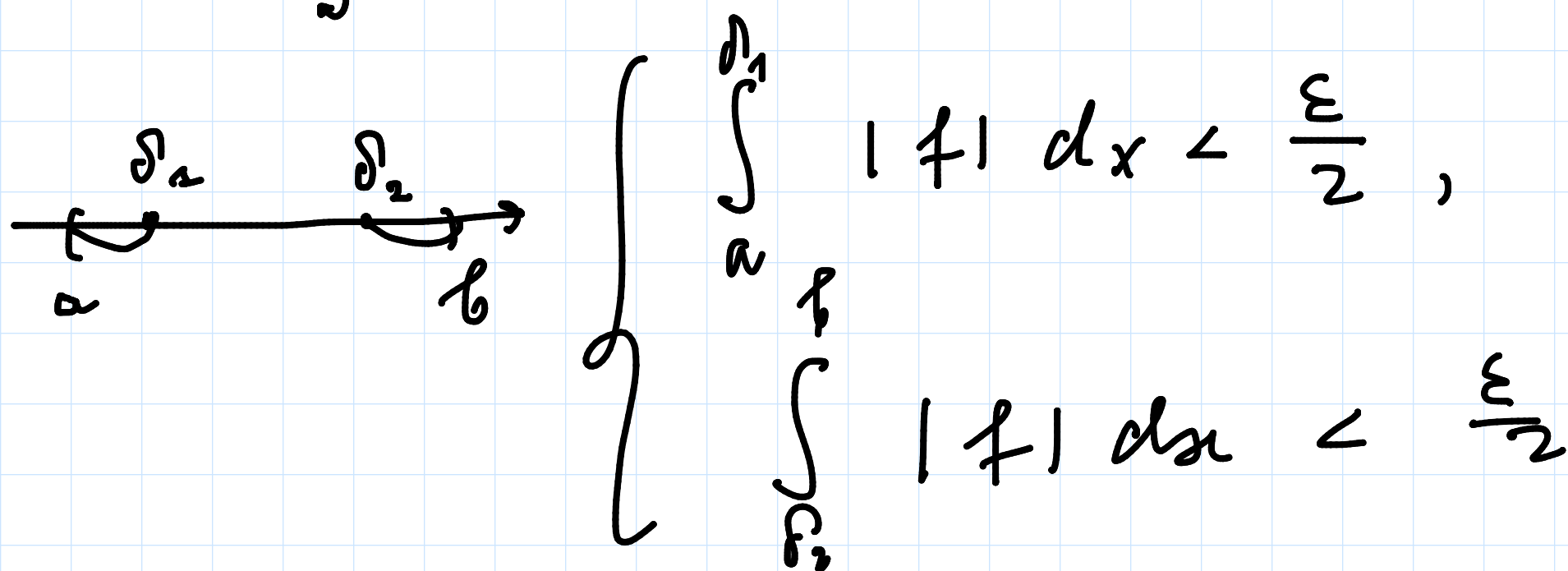
Δ 1)] $f(x) = c = \text{const}$, (a, b) - отрезок.

$$\int_a^b c e^{i\lambda x} dx = c \left(\int_a^b \cos \lambda x dx + i \int_a^b \sin \lambda x dx \right) = c \left(\frac{\sin \lambda b - \sin \lambda a}{\lambda} - i \frac{\cos \lambda b - \cos \lambda a}{\lambda} \right) \xrightarrow{\lambda \rightarrow +\infty} 0$$

2) ϕ -м: $\forall \varepsilon > 0 \exists [\delta_1, \delta_2] \subset [a, b]$:

$$\left| \int_a^b f(x) e^{i\lambda x} dx - \int_{\delta_1}^{\delta_2} f(x) e^{i\lambda x} dx \right| < \varepsilon, \forall \lambda \in \mathbb{R}$$

Т.к. $\int_a^b |f| dx < +\infty \Rightarrow \exists \delta_1, \delta_2: \delta_1 < \delta_2$:



$$\int_a^{\delta_1} |f| dx < \frac{\varepsilon}{2}, \quad \int_{\delta_2}^b |f| dx < \frac{\varepsilon}{2}$$

$$\begin{aligned} &\Rightarrow \left| \int_a^b f(x) e^{i\lambda x} dx - \int_{\delta_1}^{\delta_2} f(x) e^{i\lambda x} dx \right| = \\ &= \left| \int_a^{\delta_1} f(x) e^{i\lambda x} dx + \int_{\delta_2}^b f(x) e^{i\lambda x} dx \right| \leq \\ &\leq \int_a^{\delta_1} |f(x)| |e^{i\lambda x}| dx + \int_{\delta_2}^b |f(x)| |e^{i\lambda x}| dx = \\ &= \int_a^{\delta_1} |f(x)| dx + \int_{\delta_2}^b |f(x)| dx < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

$$\text{T.K. } f \in R [a_1, a_2] \Rightarrow \exists \tau [a_1, a_2] :$$

$$0 \leq \int_{a_1}^{a_2} f dx - \sum_{i=1}^n m_i \Delta x_i < \varepsilon !$$

$= S_\varepsilon(f)$

$$\exists g(x) = m_i,$$

$$\begin{aligned} & \left| \int_{a_1}^{a_2} f(x) e^{i\lambda x} dx - \int_{a_1}^{a_2} g(x) e^{i\lambda x} dx \right| = \\ & = \left| \int_{a_1}^{a_2} (f(x) - g(x)) e^{i\lambda x} dx \right| \leq \int_{a_1}^{a_2} |f(x) - g(x)| dx = \\ & = \int_{a_1}^{a_2} f(x) dx - S_\varepsilon(f) < \varepsilon \end{aligned}$$

$$\Rightarrow 0 \leq \left| \int_{a_1}^{a_2} f e^{i\lambda x} dx - \int_{a_1}^{a_2} g e^{i\lambda x} dx \right| < \varepsilon$$

$\lambda \rightarrow \infty \uparrow$
 $\rightarrow 0$

$$\Rightarrow \lim_{|\lambda| \rightarrow \infty} \sum_{i=1}^n \int_{x_{i-1}}^{x_i} m_i e^{i\lambda x} dx \rightarrow 0$$

$n \Rightarrow \text{T.K. } \varepsilon - \forall$

$$\Rightarrow \int_{a_1}^{a_2} f e^{i\lambda x} dx \xrightarrow{\lambda \rightarrow +\infty} 0 \quad \Delta$$

$$\Rightarrow \text{Зам } C_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx \xrightarrow{k \rightarrow +\infty} 0$$

Hy Cx $\cos x$ $\sin x$ Репе Рупе

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx dx \xrightarrow{k \rightarrow \infty} 0$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx dx \xrightarrow{k \rightarrow \infty} 0$$

(1)

$$f(x): T=2\pi$$

$$T_n(x) = \frac{1}{2\pi} \int_0^\pi [f(x-t) + f(x+t)] D_n(t) dt$$

$$\begin{aligned} \Delta \quad T_n(x) &= \frac{1}{2\pi} \int_{x-\pi}^{x+\pi} f(t) D_n(x-t) dt = \left\{ x-t=p \right\} \\ &= \frac{1}{2\pi} \int_{x-\pi}^{x+\pi} f(x-p) D_n(p) dp = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-p) \underbrace{D_n(p)}_{\text{Dini}} dp \\ &= \frac{1}{2\pi} \int_0^\pi [f(x-p) + f(x+p)] D_n(p) dp \end{aligned}$$

Опр Условие Дини

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad x \in \mathbb{R}$$

(1) $f(x \pm 0)$

(2) $\int_0^\delta \left| \frac{f(x-t) - f(x-0)}{t} \right| dt < \infty \quad \forall \delta > 0$

$\int_0^\delta \left| \frac{f(x+t) - f(x+0)}{t} \right| dt < \infty \quad \forall \delta > 0$

(Т) Покажем, что предложение Пуассе.

(1) $f: T=2\pi$

(2) $f \in R[-\pi, \pi]$

$\Rightarrow$

(3) f удовлетв. в т. $x \in \mathbb{R}$ усл. Дини

$$\sum_{k=-\infty}^{+\infty} c_k(f) e^{ikx} = \frac{f(x+0) + f(x-0)}{2}$$

$$f(x) = S(x), \quad x - \text{г. непрерыв.}$$

$$\frac{f(x_0+0) + f(x_0-0)}{2} = S(x_0)$$

$$\Delta \quad T_n(x) = \frac{1}{2\pi} \int_0^\pi [f(x-t) + f(x+t)] D_n(t) dt$$

$$\begin{aligned} \circ \quad T_n(x) - \frac{f(x+0) + f(x-0)}{2} &= \frac{1}{2\pi} \int_0^\pi [f(x-t) + \\ &+ f(x+t)] D_n(t) dt - \frac{1}{2\pi} \int_0^\pi [f(x+0) - f(x-0)] \phi_n(t) dt \\ \left\{ \int_0^\pi D_n(t) dt = \pi \right\} &\Leftrightarrow \frac{1}{2\pi} \int_0^\pi [f(x-t) - f(x-0) + \\ &+ f(x+t) - f(x+0)] \frac{\sin(n+\frac{1}{2})t}{\sin \frac{t}{2}} dt \end{aligned}$$

$$\circ \quad \int_0^\pi \left| \frac{f(x-t) - f(x-0)}{\sin \frac{t}{2}} \right| dt = C_x \quad \left(\frac{\sin \frac{t}{2}}{\frac{t}{2}} \sim \frac{t}{2}, t \rightarrow 0 \right)$$

$$\int_0^\pi \left| \frac{f(x+t) - f(x+0)}{\sin \frac{t}{2}} \right| dt = C_x$$

$\Rightarrow$ Римана

$$\lim_{n \rightarrow \infty} T_n(x) = \frac{f(x+0) + f(x-0)}{2} \quad \Delta$$

Сл. $\int$ Кус - непрерыв
Кус - монот

ггг. ггг. Дини.

$f: [-e, e]$, f непрерывна на $\mathbb{R}$

$$y = \frac{\pi}{e} x \in [-\pi, \pi]$$

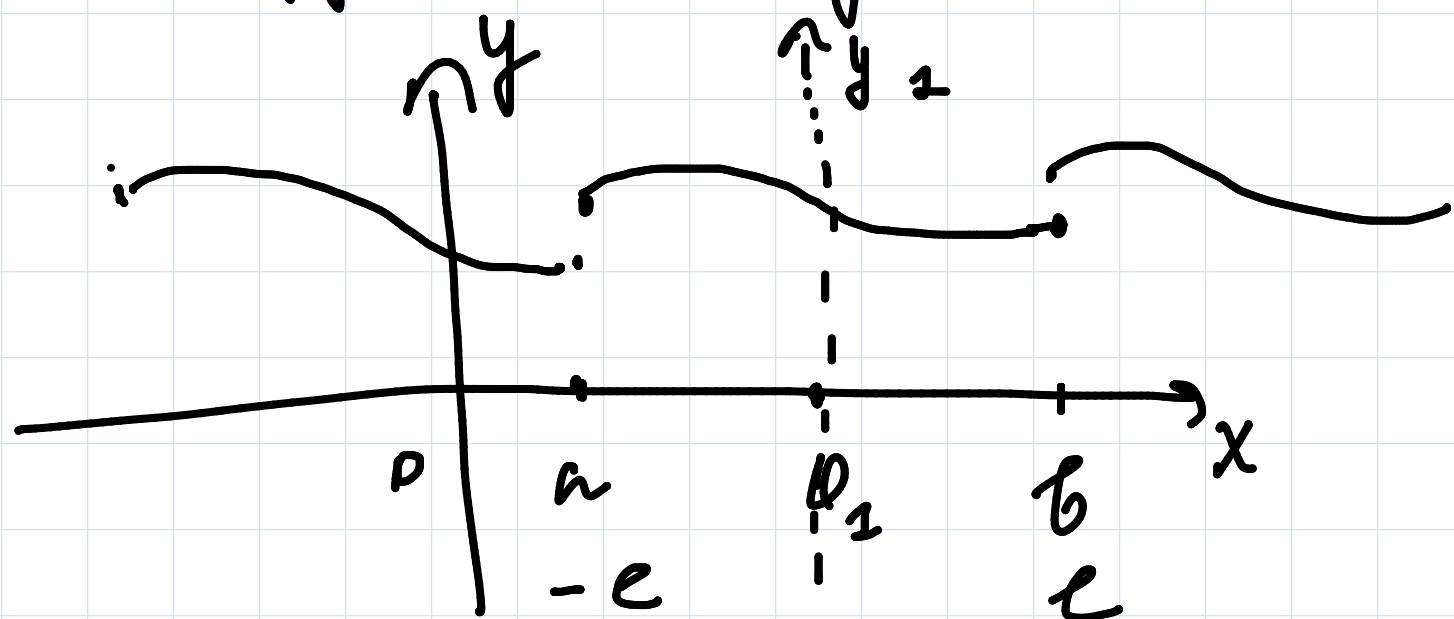
$$f \sim \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos \frac{\pi k x}{e} + b_k \sin \frac{\pi k x}{e}$$

$$\left\{ \begin{aligned} a_k &= \frac{1}{e} \int_{-e}^e f \cos \frac{\pi k x}{e} dx, \quad k=0, 1, \dots \\ b_k &= \frac{1}{e} \int_{-e}^e f \sin \frac{\pi k x}{e} dx. \end{aligned} \right.$$

$y = f(x)$ — непрерывна, задана на $\mathbb{R}$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

может быть пр. φ на $\forall [a, b]$, на кот. упр. соотв. уст.

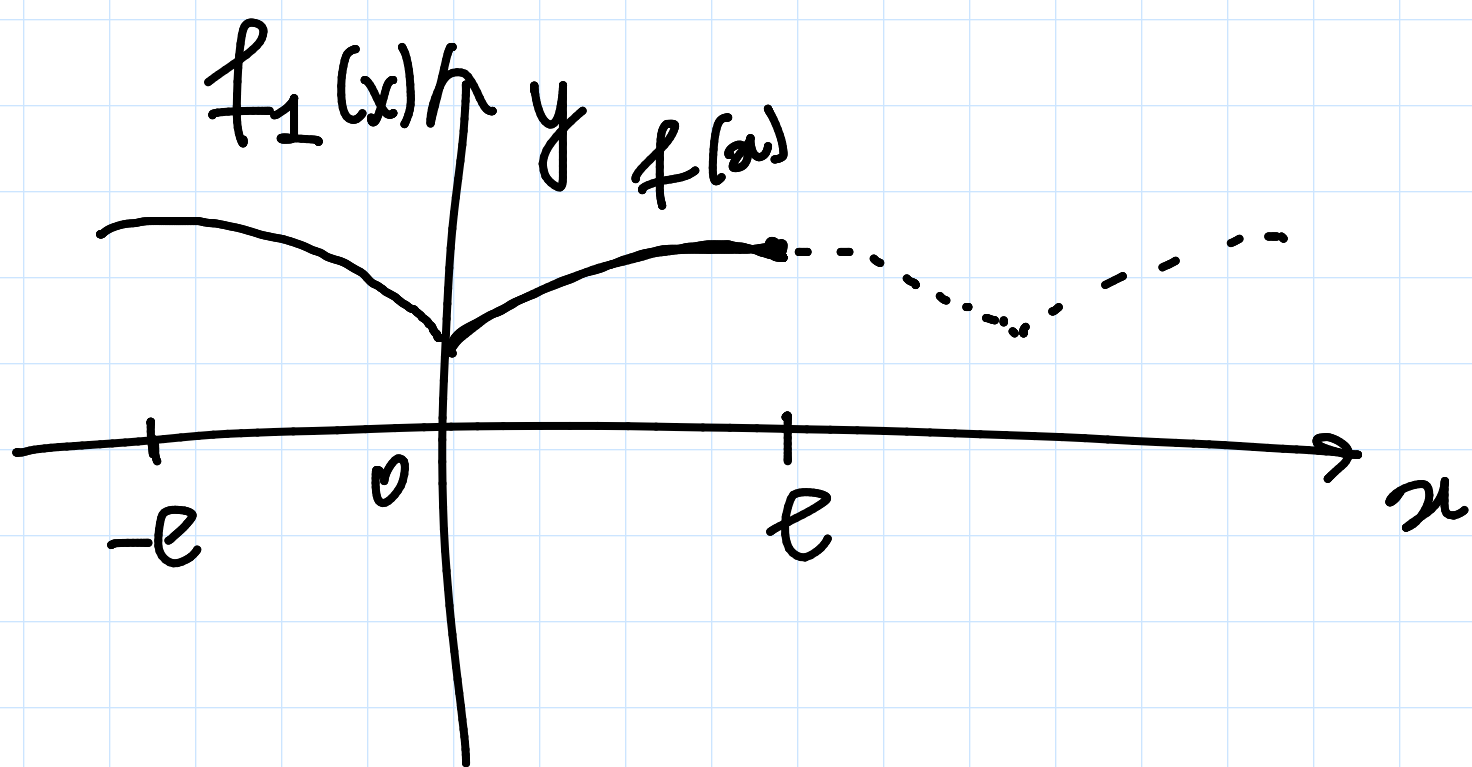


$$\varphi f_1(x) = f(x), \quad x \in [-e, e]$$

$$\tau = 2e = |b - a|$$

$f(x)$ на $[0, e]$

$$\varphi f_1(x) = \begin{cases} f(x), & x \in [0, e] \\ f(-x), & x \in [-e, 0] \end{cases}$$



$$f_1(x) = \begin{cases} f(x), & x \in [0, e] \\ -f(-x), & x \in [-e, 0] \end{cases}$$

